# Tonal noise of a ducted desk fan with serrated blades: sliding-mesh URANS, Ffowcs Williams–Hawkings prediction, and a consistency check of the acoustic output

Clément Dumeril*

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**Abstract**

This report describes a computational-aeroacoustics chain for a five-blade ducted desk fan whose trailing edges carry owl-inspired serrations, and an audit of its acoustic output. The incompressible flow is solved with OpenFOAM (URANS, $k$–$\omega$ SST) on a 1.95-million-cell hex-dominant mesh, with the rotor in a cyclicAMI sliding zone turning at 158 rad/s (1 509 rpm). The Farassat 1A form of the Ffowcs Williams–Hawkings (FW-H) equation propagates the blade-surface pressure to four microphones at 1 m. The established regime gives 43.3 mN of thrust, 1.02 mN m of torque and 108 $\mathrm{m}^3$/h. The spectra show the blade-passing frequency (BPF, 126 Hz) 20 to 28 dB above the local floor on every microphone, and a once-per-revolution tone that exists only in the rotor plane, with the $\cos\theta$ directivity of a compact dipole rotating with the rotor; the blade forces contain the matching rotating in-plane component (0.95 mN). Re-processing the two 85 ms FW-H records separately removed two errors of the first analysis (a concatenation across a 15 ms gap and a median filter that raised the floor by 9 to 24 dB). A compact-dipole estimate built from the integrated forces does not reproduce the FW-H levels (+12 dB at 25 Hz, −18 dB at the BPF), so only tone frequencies and directivity are treated as results. No mesh or time-step independence study was run, and the effect of the serrations is not measured.

## 1 Introduction

The tonal noise of a low-speed fan is radiated by the unsteady force the blades exert on the air. In the stationary frame, a rotor of $Z$ identical blades turning at $n$ revolutions per second radiates at the blade-passing frequency $Zn$ and its harmonics; any asymmetry between blades adds tones at multiples of $n$. Trailing-edge serrations, inspired by owl feathers, act mainly on the broadband part of the spectrum by reducing the scattering of boundary-layer turbulence at the edge [1, 2]. A recent study combined owl-feather serrations with cicada-wing geometry on small propellers and reported lower overall levels and higher propulsive efficiency [3]; that work was the starting point of the fan studied here.

The question of this study was practical: can an open-source chain, run on a small self-built cluster, give a usable acoustic signature of a 3D-printable desk fan before it is built, and which parts of that signature can be trusted? The fan has five serrated blades in a short duct, with a motor hub held by three arms.

The contributions are:

*University of California, Berkeley (MEng). Work carried out in May–June 2026 at Polytechnia, an engineering consultancy co-founded by the author; re-analysis of the acoustic data in October 2026. Repository: `https://github.com/clementdumeril/ducted-fan-aeroacoustics`.

- a sliding-mesh URANS and FW-H set-up for the fan (OpenFOAM, libAcoustics), with the failed non-conformal-coupling attempt and the fixes that made the cyclicAMI run stable;
- a per-record spectral analysis of the FW-H signals, which corrects two errors of the first analysis and identifies a once-per-revolution tone whose directivity matches a rotating in-plane force;
- a consistency check of the FW-H levels against the integrated blade forces, which shows that the absolute tone levels of this run are not reliable.

# 2 Related work

The Ffowcs Williams–Hawkings equation [4] extends Lighthill's analogy to sources on moving surfaces. For surfaces moving at subsonic speed, Farassat's Formulation 1A gives the thickness and loading noise as retarded-time surface integrals with the observer-time derivative taken analytically [5]; it is the standard tool for rotor tonal noise [6]. A 2026 preprint argues that Formulation 1A omits a term linked to the variation of the surface normal, which matters for surfaces that do not move in pure translation [7]; this was not examined here. The FW-H implementation used is libAcoustics [8, 9], a library of function objects for OpenFOAM [10, 11]. Rotor motion uses the arbitrary mesh interface (AMI), whose conservative interpolation between non-matching meshes follows the local Galerkin projection of Farrell and Maddison [12]. The flow uses the $k$–$\omega$ SST model [13]. On serrations, Howe [1] gave the first analytical model of sawtooth trailing-edge noise and Lyu et al. [2] a more complete prediction; both concern broadband noise, which a URANS simulation does not resolve. Spectral estimation follows the windowing practice of Harris [14]; averaging periodograms over segments is Welch's method [15].

# 3 Model

## 3.1 Geometry and operating point

The geometry is the Polytechnia FreeCAD model of the fan, exported to STL: five-blade rotor (tip radius 69.2 mm, serrated trailing edges), duct of inner radius 76 mm, motor hub and three support arms. Table 1 lists the main parameters. The rotor turns at $\Omega = 158\,\mathrm{rad/s}$, that is 1 509 rpm or $n = 25.15\,\mathrm{Hz}$, so BPF $= 5n = 125.7\,\mathrm{Hz}$. The tip speed is $\Omega R = 10.9\,\mathrm{m/s}$ (tip Mach number 0.03): the flow is incompressible and the rotor is acoustically compact at the BPF ($kR = 0.16$).

**Table 1:** Model parameters (files in `openfoam_case/`).

| Quantity | Value |
|---|---|
| Rotor | 5 blades, tip radius 69.2 mm, serrated trailing edges |
| Rotation | $\Omega = 158\,\mathrm{rad/s}$ (1 509 rpm), ramp from 0 over 5 ms |
| Fluid | air, $\rho = 1.2\,\mathrm{kg/m^3}$, $\nu = 1.5 \times 10^{-5}\,\mathrm{m^2/s}$ |
| Domain | box $0.4\,\mathrm{m} \times 0.4\,\mathrm{m} \times 0.5\,\mathrm{m}$, axis along $+z$ |
| Open boundaries | total pressure $p_0 = 0$, `pressureInletOutletVelocity` |
| Walls | no slip; blades `movingWallVelocity`; wall functions for $k$, $\omega$ |
| Inflow turbulence | $k = 0.54\,\mathrm{m^2/s^2}$, $\omega = 128\,\mathrm{s^{-1}}$ |
| Observers | $r = 1\,\mathrm{m}$: downstream and upstream on the axis, radial, 45° downstream |

### 3.2 Equations

The incompressible Navier–Stokes equations are solved with the $k$–$\omega$ SST closure [13]. The rotor zone turns as a rigid body; fluxes cross the cylindrical interface ($R = 72\,\mathrm{mm}$) through the AMI. The acoustic pressure at observer $\mathbf{x}$ is obtained from the FW-H equation with the blade skin as the integration surface, keeping the thickness and loading terms and neglecting the quadrupole term:

$$4\pi p'_T(\mathbf{x},t) = \int_S \left[\frac{\rho_0(\dot{v}_n + v_{\dot{n}})}{r(1-M_r)^2}\right]_{\mathrm{ret}} \mathrm{d}S + \int_S \left[\frac{\rho_0 v_n\,(r\dot{M}_r + c_0 M_r - c_0 M^2)}{r^2(1-M_r)^3}\right]_{\mathrm{ret}} \mathrm{d}S, \tag{1}$$

$$4\pi p'_L(\mathbf{x},t) = \frac{1}{c_0}\int_S \left[\frac{\dot{\ell}_r}{r(1-M_r)^2}\right]_{\mathrm{ret}} \mathrm{d}S + \int_S \left[\frac{\ell_r - \ell_M}{r^2(1-M_r)^2}\right]_{\mathrm{ret}} \mathrm{d}S + \frac{1}{c_0}\int_S \left[\frac{\ell_r\,(r\dot{M}_r + c_0 M_r - c_0 M^2)}{r^2(1-M_r)^3}\right]_{\mathrm{ret}} \mathrm{d}S, \tag{2}$$

with $p' = p'_T + p'_L$, in the notation of Farassat [5] and Brentner and Farassat [6]: $v_n$ is the normal surface velocity, $\ell_r$ and $\ell_M$ are the surface loading projected on the radiation direction and on the surface Mach number vector, $M_r$ is the Mach number in the radiation direction, and dots denote derivatives with respect to source time. Because the solver is incompressible, the surface pressure is a kinematic pressure; the FW-H input uses $\rho_0 = 1.2\,\mathrm{kg/m^3}$ and $c_0 = 343\,\mathrm{m/s}$.

## 4 Numerical set-up

### 4.1 Mesh

The mesh is built with `snappyHexMesh` from a 10 mm background grid, refined to level 5 (0.31 mm) on the blades so that the serrations are resolved, level 3 on the duct and arms and level 2 on the hub, with a refined rotor cylinder and wake region (Figure 1). The faceZone of the rotor cylinder is split into a cyclicAMI pair with `createBaffles`. The retained mesh has 1 949 584 cells; its quality is listed in Table 2. The intended low-Reynolds prism layers could not be inserted on the thin serrated blades, so the blade $y^+$ averages about 4 and reaches 445.

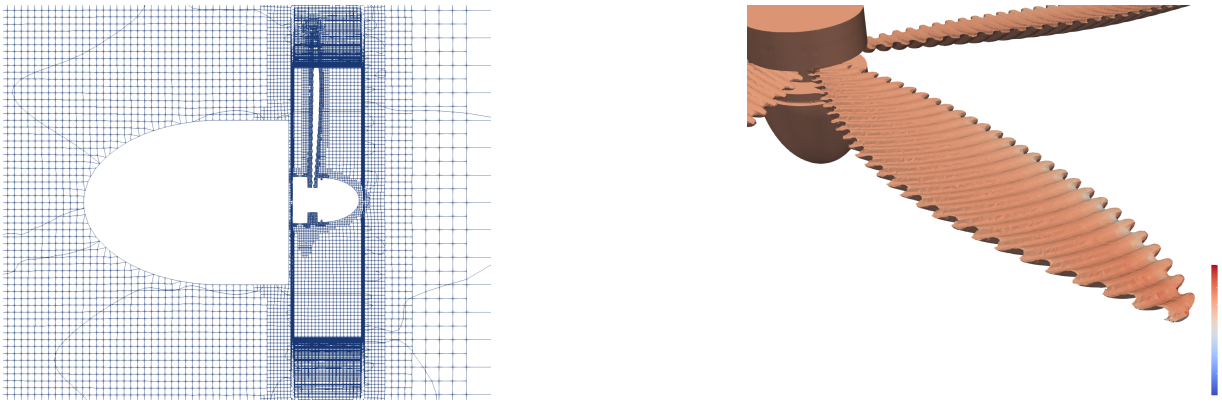


**Figure 1:** Left: meridional cut of the mesh, from the far field to the blade band and the AMI cylinder. Right: serrated trailing edges as resolved by the mesh.

**Table 2:** `checkMesh` on the retained mesh.

| | |
|---|---|
| Cells / points | 1 949 584 / 2 205 401 |
| Max aspect ratio | 6.61 |
| Max / mean non-orthogonality | 61.4 / 8.2 |
| Max skewness | 7.25 (12 faces above 4) |
| Concave cells | 49 827 (2.6 %) |
| Cells with determinant $< 0.001$ | 1851 |
| AMI weights (min / mean / max) | 0.90 / 1.002 / 1.27 |

### 4.2 Solver and time strategy

`pimpleFoam` is run with three outer correctors, two pressure correctors and one non-orthogonal corrector. Convection of momentum uses `linearUpwindV`; $k$ and $\omega$ use `limitedLinear`; time integration is first-order implicit Euler. The run has two phases. Phase 1, from rest to 0.19 s (about 4.8 revolutions), uses an adaptive time step limited by a Courant number of 3 and $\Delta t \leq 1 \times 10^{-4}$ s, with the FW-H function object off. Phase 2 restarts with a fixed $\Delta t = 3.5 \times 10^{-5}$ s and the FW-H function object on, and runs to 0.39 s. The fixed step is confirmed by the force output, which is written every 10 steps at a uniform interval of $3.5 \times 10^{-4}$ s. The FW-H pressure is sampled every 5 steps ($f_s = 5\,714$ Hz) and written in two records of 100 ms.

The degenerate cells make the solver stop with a floating-point exception at the first step unless `FOAM_SIGFPE=false` is set; this tolerates them without changing their quality.

### 4.3 How the set-up was reached

Steady multiple-reference-frame runs at 3 000 and then 4 320 rpm (end of May 2026) gave first flow rates. The first transient attempt used the non-conformal coupling of OpenFOAM.org: over six mesh variants (re-ordered interfaces, Delaunay and rounded end caps of the rotor cylinder) the runs stopped in the mesh stitcher after about 200 µs of simulated time. The case was then moved to the ESI OpenFOAM release with cyclicAMI (12 June 2026). Two further changes were needed: meshing in serial, because the parallel `snappyHexMesh` aborted while load-balancing the AMI faceZone, and a PCG solver instead of GAMG for the pressure on the coarse test mesh, where a zero-determinant cell made GAMG ill-conditioned.

### 4.4 Acoustic post-processing

Each FW-H record begins with a start-up transient: on the axial microphones its peak reaches 35 to 52 robust standard deviations of the rest of the record. The first 15 ms of each record are discarded, which leaves two segments of 85 ms (486 samples) separated by a 15 ms gap (sim. time 0.205 s to 0.290 s and 0.305 s to 0.390 s). Each segment has its mean and linear trend removed, is Hann-windowed and zero-padded to 8192 points; the two power spectra are averaged. The true frequency resolution is 11.8 Hz. Tone levels are the spectrum peaks within ±6 Hz of $n$ and of the BPF; the local floor is the median level between 140 and 240 Hz.

### 4.5 Computing

The runs used a two-node Linux cluster built for this project (8 and 4 cores, MPI over Ethernet, shared NFS directory). A `simpleFoam` benchmark on 782 400 cells showed that one node saturates at 4 cores (1.51 times faster than 1 core; 1.44 times on 8 cores) and that 12 cores on both nodes

reach 1.93 times. For the AMI case, splitting one run across the two nodes gave no gain, so the cluster was used to run independent cases side by side. A logged stretch of phase 1 (961 steps) took 15 844 s of CPU time, about 16.5 s per step.

## 5 Verification

**Established regime.** The thrust varies by less than 1 % from one revolution to the next after about 4.75 revolutions, before the FW-H recording starts (Figure 2).

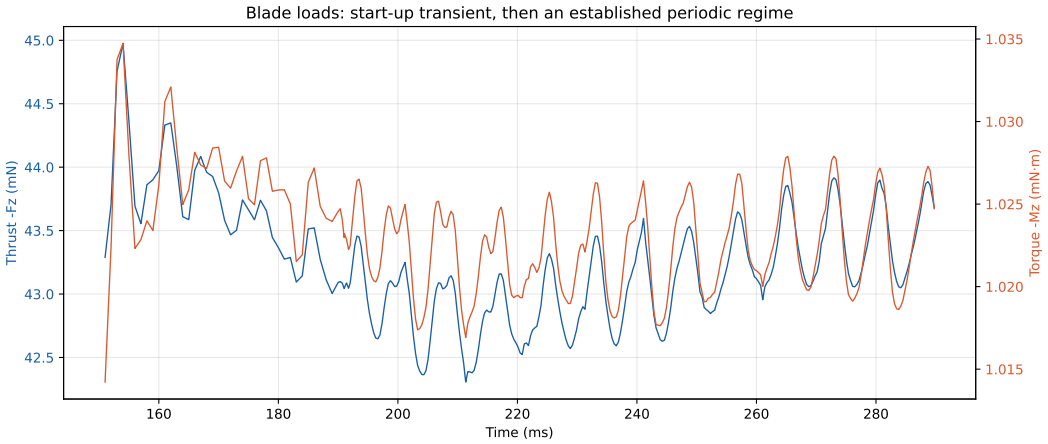



**Figure 2:** Thrust and torque on the blades: start-up transient, then a periodic regime.

**Post-processing audit.** The first analysis of these records concatenated the two segments and applied a 5-point median filter against numerical spikes. Both choices were re-examined. The concatenation joins two segments 15 ms apart, which puts a phase jump of $360° \times 15\,\text{ms} \times 25.15\,\text{Hz} \approx 136°$ into the once-per-revolution component and splits its peak. The segments contain few outlier samples (at most 4 beyond 5 robust standard deviations), and the filter raises the floor by 9 to 24 dB, hides the BPF harmonics and moves the radial BPF level by about 4 dB (Figure 3). The results below use the per-segment, unfiltered processing of Section efsec:post.

**Coherence between records.** On the radial and 45° microphones, a sine fit at 25.15 Hz on the absolute time axis gives the same level (21.0 and 17.9 dB) and the same phase (148°) in both segments: the once-per-revolution tone is a steady feature of the run. On the axial microphones the fitted phase changes from one segment to the other and the level stays at the floor.

**Mesh quality.** See Table 2. The effect of the tolerated cells on the spectra was not measured.

**Not verified.** No mesh-independence study (one mesh retained), no second time step, no resolved boundary layer on the blades, and no comparison with a measurement.

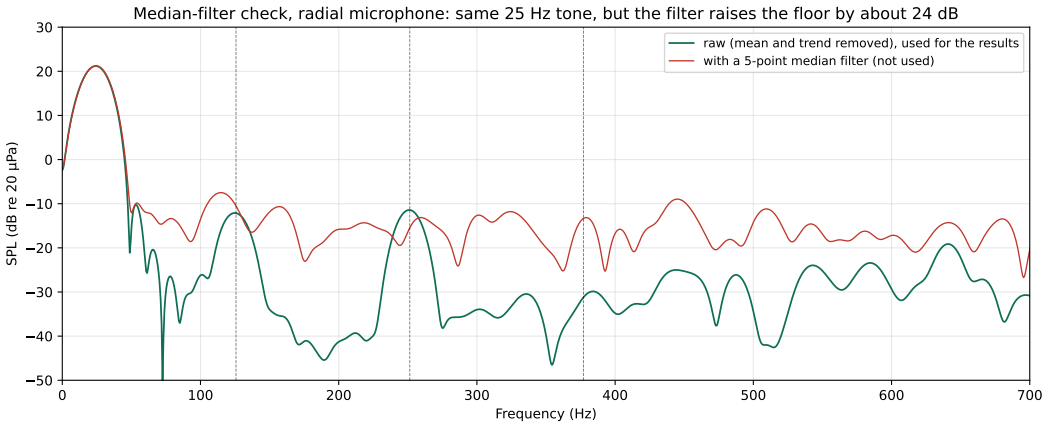



**Figure 3:** Radial microphone, average of the two segments: raw signal (used) and 5-point median filter (not used).

# 6 Results

## 6.1 Aerodynamics

Averaged over the last 50 ms of the force record, the rotor gives 43.3 mN of thrust and 1.023 mN m of torque, that is an aerodynamic power $C\Omega = 0.162$ W. The flow rate through a cut 60 mm downstream of the rotor is 30.1 L/s (108 $m^3$/h). The blades drive an annular jet along the duct wall, with a slow recirculating wake behind the hub (Figure 4).

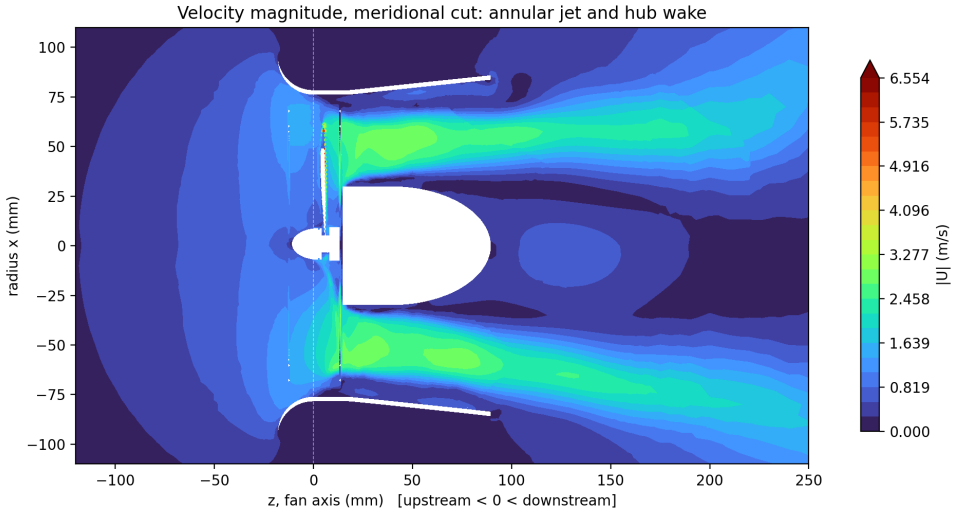



**Figure 4:** Velocity magnitude in the meridional plane at the last time step.

## 6.2 Spectra and directivity

Figure 5 shows the averaged spectra and Table 3 the tone levels. The BPF stands 20 to 28 dB above the local floor on all four microphones. Its second harmonic appears on every microphone and its third on all but the radial one. The once-per-revolution tone dominates in the rotor plane and is absent on the axis.

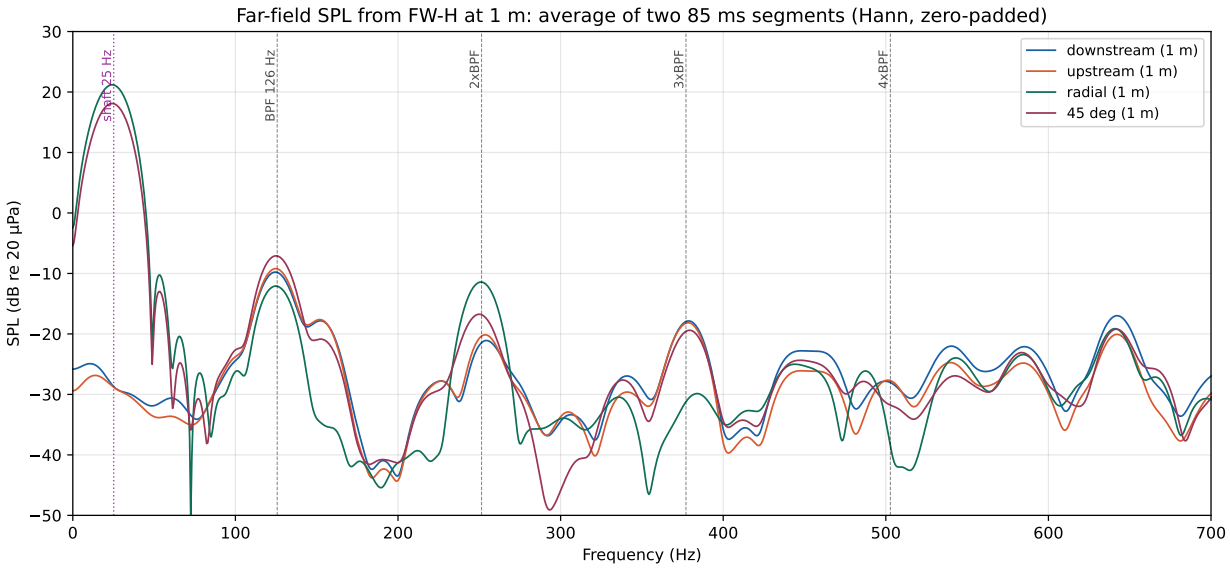



**Figure 5:** Far-field spectra at 1 m: average of the two 85 ms segments, Hann window, raw signal.

**Table 3:** Tone levels at 1 m (dB re 20 µPa, raw signal). Absolute levels are not considered reliable (Section efsec:levels); only comparisons between microphones and tone-to-floor ratios are used.

| Microphone | 25 Hz | BPF | floor (140–240 Hz) | BPF − floor |
|---|---|---|---|---|
| Downstream (axis) | −26.7 (floor) | −9.8 | −30.0 | 20.3 |
| Upstream (axis) | −27.5 (floor) | −9.2 | −30.1 | 20.9 |
| Radial | 21.2 | −12.1 | −39.8 | 27.7 |
| 45° downstream | 18.1 | −7.1 | −30.1 | 23.0 |

## 6.3 Origin of the once-per-revolution tone, and a check of the levels

The net force on the blades has an in-plane component of 0.95 mN that rotates with the rotor: $F_x$ and $F_y$ oscillate at 25.15 Hz with amplitudes 0.945 and 0.948 mN. At 25 Hz the rotor is compact (wavelength 13.6 m), and a compact dipole rotating in the rotor plane radiates nothing on the axis and $\cos 45°$ of its in-plane amplitude at 45°, that is −3.0 dB. The FW-H data give no tone on the axis and a 45° level 3.1 dB below the radial one. The directivity of the tone is therefore that of the rotating in-plane force. Whether that force is physical (a blade-to-blade difference of the printed geometry) or a mesh effect was not established.

The levels do not agree. Table 4 compares the FW-H levels with the compact-dipole estimate $|p'| = A\,(4\pi r)^{-1}\sqrt{(\omega/c_0)^2 + r^{-2}}$ computed from the force amplitudes $A$ over the same time window (script `paper/scripts/dipole_check.py`). FW-H is 12 dB above the estimate at 25 Hz and 18 dB below it at the BPF on the axis. The two discrepancies have opposite signs, so a single unit factor (such as the kinematic-pressure convention) cannot explain both. Candidate causes are the treatment of the rotating integration surface in the FW-H function object and the omitted surface-normal term discussed by [7]; neither was tested. Until this is resolved, the tone frequencies, their presence on each microphone and the 25-Hz directivity are the results of this study, not the absolute levels.

**Table 4:** FW-H levels against a compact-dipole estimate from the integrated blade forces ($r = 1\,\mathrm{m}$).

| Tone, microphone | force amplitude (mN) | compact dipole (dB) | FW-H (dB) |
|---|---|---|---|
| 25 Hz, radial (+x) | 0.94 | 9.3 | 21.2 |
| 25 Hz, axis | – | no radiation | at floor |
| BPF (124.5 Hz), downstream axis | 0.36 | 8.1 | -9.8 |

# 7 Discussion and limitations

The chain produces a physically consistent flow field and a spectrum with the expected structure: BPF and harmonics everywhere, plus a once-per-revolution tone whose directivity is explained by the measured force history. The quantitative acoustic results are weaker than the aerodynamic ones, for five reasons.

- *Inconsistent levels.* The FW-H levels disagree with the force-based estimate in both directions (Table 4).
- *Resolution.* Two 85 ms segments give an 11.8 Hz resolution; the once-per-revolution tone at 25 Hz sits only two bins from zero. A contiguous 0.4 s record would need about 11 400 steps, roughly 55 to 60 h at the measured cost, and the cluster that holds the restart data was offline when this report was written.
- *Numerical accuracy.* One mesh, one time step, first-order time integration at Courant numbers up to 3 in phase 1, no prism layers on the blades, and 49 827 concave cells.
- *Physics.* URANS resolves tonal noise only. The serrations act mainly on broadband noise [1, 2], and no smooth-blade reference was run, so this study does not measure their effect. The FW-H surface covers the blades only; the duct, arms and hub are not part of it.
- *No measurement.* Thrust, flow rate and spectra are not compared with tests.

# 8 Conclusion and future work

A sliding-mesh URANS and FW-H chain was set up for a serrated ducted fan and run on a two-node cluster; it gives 43.3 mN of thrust and 108 $\mathrm{m}^3$/h at 1 509 rpm, a BPF tone with harmonics on all microphones and a once-per-revolution tone confined to the rotor plane. Re-processing the FW-H records per segment corrected two analysis errors, and a force-based compact-dipole check showed that the absolute FW-H levels of this run are not reliable. In order of priority, the next steps are: find the cause of the level discrepancy on a case with a known answer (a rotating compact dipole, or a single rotating blade); record a contiguous signal of at least 0.4 s; run two more meshes and a second time step; add prism layers; and run a smooth-blade reference rotor under the same conditions before drawing any conclusion on the serrations.

# Reproducibility statement

Case files: `openfoam_case/` (phase 1 `system/controlDict`, phase 2 `system/controlDict.phase2`); the STL geometry is not published. Solver: OpenFOAM (ESI; v2506 on the cluster, v2406 on the machine that launched phase 2) with libAcoustics. Post-processing, from the repository root,

in Python 3 with numpy, scipy and matplotlib (each script runs in seconds): `postprocessing/acoustics/post_acoustique_combine.py` (spectra, Table 3, Figures 3 and 5), `postprocessing/aero/perf_aero.py` (thrust, torque, power, Figure 2), `paper/scripts/dipole_check.py` (Table 4). The flow rate and Figure 4 need ParaView slices of the last time step that are not published (`postprocessing/rendering/render_flow_2d.py`). The paper is rebuilt with `paper/build_paper.sh`.

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